

Single Particle Representation

$$\psi_p(r, t) = \sum_j c_j(t) \phi_j(r, R)$$

Time Dependent Kohn-Sham Equation

$$i\hbar \frac{\partial \psi_p(r, t)}{\partial t} = \mathcal{H}(r; R) \psi_p(r, t)$$

Non-Adiabatic Coupling

$$T_{jk} = \langle \phi_j | \frac{\partial}{\partial t} | \phi_k \rangle$$

Coefficient Propagation

$$i\hbar \dot{c}_j(t) = \sum_k c_k(t) (\epsilon_k \delta_{jk} - i\hbar T_{jk} + H_{jk}^{\text{LMI}})$$

Light-Matter Interaction

$$H_{jk}^{\text{LMI}} = \frac{e}{m} \mathbf{A}(t) \langle \phi_j | e \cdot \mathbf{p} | \phi_k \rangle$$

Fewest Switches Surface Hopping

$$P_{jk}(t, \Delta t) = \max \left[-\frac{2 \int_t^{t+\Delta t} \Re(\rho_{jk} H_{jk}^{\text{LMI}} / i\hbar - \rho_{jk} T_{jk}) dt}{\rho_{jj}}, 0 \right]$$

Classical Path Approximation

Detailed Balance Factor

$$G_{jk} = P_{jk} B_{jk}; \quad B_{jk} = \begin{cases} e^{-(\epsilon_j - \epsilon_k / k_B T)} & \epsilon_j > \epsilon_k \text{ and } |E| = 0 \\ 1 & \epsilon_j \leq \epsilon_k \text{ or } |E| > 0 \end{cases}$$

Result Analysis